FragTree

Predicting EI-MS Spectra of TMS-Derivatized Metabolites

using a Bipartite Fragmentation Graph and Graph Neural Networks

John Feczko | Case Western Reserve University | 2026

The Problem

Unknown annotations

Untargeted metabolomics generates thousands of spectra with no library match. Predicted spectra fill the gap.

TMS compounds underserved

CFM-ID and MassFormer were not designed for derivatized compounds. TMS chemistry has distinct, predictable fragmentation patterns.

Rare or unstable molecules

Many metabolites cannot be purchased as standards. A generative model enables library augmentation without synthesis.

Goal: train a model that scores every bond and predicts which break, and with what frequency, under EI conditions.

Reference Database

Curating a TMS-Specific Library from NIST23

1. **Export from NIST23**
   All .msp files via Lib2NIST
2. **Fragment gate**
   Requires >= 2 of: m/z 73, 147, 149
3. **Name gate**
   'tms' or 'trimethylsilyl' in name
4. **Store**
   SQLite metadata + gzip HDF5 spectra

32,866
TMS spectra retained

30,535
unique compounds (CAS)

369 Da
mean molecular weight

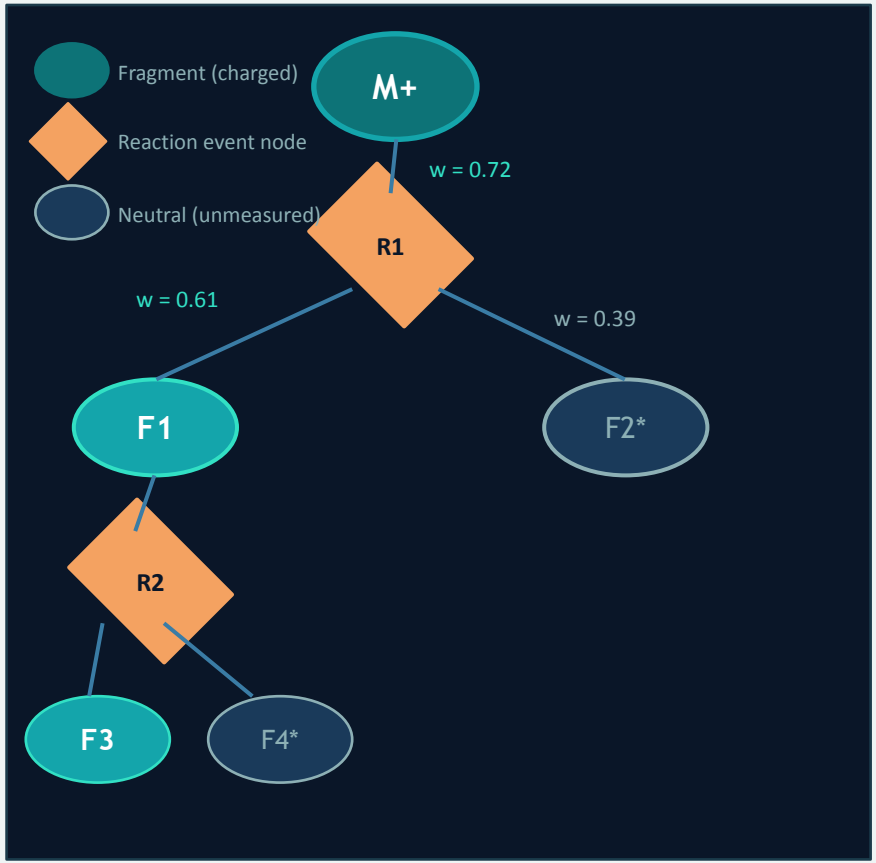
0.26
mean pairwise cosine sim.

Fragment gate first: any genuine TMS compound must produce m/z 73. Name gate removes false positives that carry those fragments for unrelated chemical reasons.

The FragTree Graph Model

02

Weighted Directed Acyclic Bipartite Graph



Directed Acyclic

Enforces one-way fragmentation flow. Prevents cycles that violate conservation of mass.

Bipartite structure

Fragment nodes connect only through reaction nodes. A single event naturally produces two children (one charged, one neutral).

Edge weights = probabilities

P(child | parent) on each edge. P(fragment) = sum over all paths of the product of edge weights along each path.

Min mass threshold: 100 Da

Aggressively prune small universal fragments. Keeps graph tractable and removes low-information peaks.

How Fragment Probabilities Are Computed

1

Edge weight

```
w(e) = P( reaction | parent fragment )
```

Predicted by a GNN operating on the molecular graph of the parent fragment

2

Path probability

```
P(path) = product of w(e) along path from root to fragment
```

Treat each fragmentation step as conditionally independent

3

Fragment probability

```
P(fragment) = sum of P(path) over all paths to that fragment
```

Sum every possible route from the molecular ion to this fragment

4

Predicted spectrum

```
I(fragment) = P(fragment) / max[ P(all fragments) ]
```

Normalize to the most probable fragment, directly comparable to experimental spectra

Comparison to experiment uses cosine similarity between predicted I(fragment) vector and measured spectrum.

Graph Neural Network Edge Weight Prediction

03

Learning Bond Cleavage Probabilities with a GNN

Input Features

Atom features

- Atomic number
- Formal charge
- Degree / valence
- Hybridization state
- Aromaticity
- Radical status

Bond features

- Bond order (1, 2, 3)
- Conjugation
- Ring membership
- Proximity to radical center

Radical proximity is the strongest predictor of bond lability -- explicitly encoded as a bond feature.

Architecture

Molecular graph of parent fragment

↓

Message passing layers (k rounds)

↓

Node embeddings per atom

↓

Bond embedding = concat(atom_i, atom_j)

↓

MLP + sigmoid => P(bond cleaves)

Training signal: cosine distance between predicted and NIST experimental spectrum.

Clustering for Training Set Selection

04

Identifying a Structurally Coherent Training Set

Morgan Fingerprints
2048-bit ECFP
RDKit, radius 2

→

UMAP Reduction
Preserves local + global structure

→

HDBSCAN Clustering
Density-based, no k required

→

Select Tightest Cluster
Highest DBCV + persistence score

Cluster Quality Metrics

Metric	What it measures	Target
DBCV	Primary tuning metric. Designed for density-based methods.	**Maximize**
Cluster Persistence	Per-cluster stability from HDBSCAN condensed tree. Confidence score per cluster.	**Maximize**
Silhouette Score	Mean intra-cluster cohesion vs inter-cluster separation.	**Maximize**
Noise Fraction	Proportion of spectra labeled as noise (-1). Too high = min_cluster_size too large.	**< 30%**
ARI + AMI	Agreement with CAS number groupings as soft ground truth.	**Maximize**

Evaluation vs. CFM-ID and MassFormer

05

Benchmarking: Competitors and Evaluation Metrics

Competing Methods

CFM-ID 4.0

Probabilistic / rule-based

Models fragmentation as a Markov process. Learns transition probabilities via expectation-maximization. Most widely used dedicated EI-MS prediction tool.

MassFormer

Transformer deep learning

Predicts spectra directly from molecular graph representations via a transformer architecture. No explicit fragmentation modeling. Current deep learning state of the art.

FragTree (ours)

Bipartite graph + GNN

Explicitly models fragmentation events as graph nodes. Trained specifically on TMS-derivatized compounds. Edge weights predicted by a bond-level GNN.

Evaluation Metrics

Cosine similarity [PRIMARY]

PRIMARY. Dot product of L2-normalized intensity vectors. Directly used in spectral library search.

Weighted cosine similarity

Weights m/z by sqrt(intensity). Reduces dominance of base peak (m/z 73) in TMS spectra.

Top-k peak recall

Fraction of top-k experimental peaks in top-k predicted peaks. Evaluated at k = 5, 10, 20.

Match rate at thresholds

Proportion of test spectra with cosine sim. > 0.7, > 0.8, > 0.9 -- standard library search thresholds.

Summary

- 32,866 TMS spectra curated from NIST23 with a two-stage fragment + name filter
- Bipartite DAG: fragment nodes through reaction event nodes -- conserves mass, handles two-product cleavage naturally
- GNN predicts bond cleavage probabilities from atom and bond features including radical proximity
- P(fragment) = sum of path probabilities from root, normalized to most probable fragment
- Benchmarked against CFM-ID 4.0 and MassFormer: cosine similarity, weighted cosine, top-k recall, match-rate thresholds

Questions?